

DEMO-101:2026

Illustrative Packaged Drinking Water Specification (synthetic demo document, not an Indian Standard)

First Revision - ICS 13.060.20

FOREWORD

This is a synthetic demonstration document created for the BISense prototype. It is not an Indian Standard, it has not been issued by the Bureau of Indian Standards, and all values in it are illustrative only. It exists so that every feature of the BISense pipeline can be exercised when no real standard has been loaded.

This illustrative specification covers water that is treated and packed in sealed containers for drinking. It was drafted in the style of a product specification, with requirements, test methods, sampling, marking and packing clauses.

In reporting the result of a test or analysis made in accordance with this document, the final value, observed or calculated, shall be rounded off in accordance with DEMO-900.

1 SCOPE

1.1 This document prescribes the requirements and the methods of sampling and test for packaged drinking water (other than packaged natural mineral water).

1.2 This document does not apply to packaged natural mineral water, which is covered in DEMO-102.

2 REFERENCES

The following documents contain provisions which, through reference in this text, constitute provisions of this document:

DEMO-102 Illustrative Packaged Natural Mineral Water Specification

DEMO-900 Illustrative Rules for Rounding Off Numerical Values

DEMO-905 Illustrative Methods of Sampling and Microbiological Examination of Water

3 TERMINOLOGY

For the purpose of this document, the following definitions shall apply.

3.1 Packaged Drinking Water - Water derived from any source of potable water which is subjected to treatment such as decantation, filtration, disinfection or reverse osmosis, and which is packed in hermetically sealed containers of various compositions, forms and capacities that are suitable for direct consumption without further treatment.

3.2 Container - Any bottle, jar, pouch or cup made of food-grade material in which the water is packed and sealed.

3.3 Batch - All containers of the same capacity packed on one line in one shift from the same source.

4 REQUIREMENTS

4.1 Source and Treatment

4.1.1 The water shall be obtained from a source that is free from contamination and shall be treated by one or more of the following methods: decantation, filtration, aeration, disinfection by ozone or ultraviolet radiation, and reverse osmosis.

4.1.2 Water shall not be treated with any substance that adds colour, odour or taste.

4.1.3 The treatment plant should be inspected at least once in every six months.

4.2 Physical and Chemical Requirements

The water shall comply with the physical and chemical requirements given in Table 1 when tested by the methods referred to in column 5 of Table 1.

Table 1 Physical and Chemical Requirements for Packaged Drinking Water

SI No.	Characteristic	Unit	Requirement	Method of Test
i	Colour, Max	Hazen units	2	Clause 8.1
ii	Odour	-	Agreeable	Clause 8.2
iii	Turbidity, Max	NTU	2	Clause 8.3
iv	Total dissolved solids, Max	mg/l	500	Clause 8.4
v	pH value	-	6.5 to 8.5	Clause 8.5
vi	Nitrate (as NO ₃), Max	mg/l	45	Clause 8.6
vii	Fluoride (as F), Max	mg/l	1.0	Clause 8.6
viii	Chloride (as Cl), Max	mg/l	200	Clause 8.6
ix	Lead (as Pb), Max	mg/l	0.01	Clause 8.6

4.3 Microbiological Requirements

4.3.1 Coliform bacteria shall be absent in any 250 ml sample of the water.

4.3.2 Escherichia coli shall be absent in any 250 ml sample of the water.

4.3.3 The total viable colony count at 37 degrees C shall not exceed 20 per ml.

4.4 Hygiene

4.4.1 The water shall be processed, packed, stored and distributed under hygienic conditions in licensed premises.

4.4.2 Containers may be reused only after cleaning and sanitation by an automatic washing process.

5 PACKING

5.1 The water shall be packed in clean, colourless, transparent and hermetically sealed food-grade containers.

5.2 Containers shall not be filled beyond 97 percent of their capacity.

5.3 Pouches should be packed in secondary cartons to protect them from direct sunlight.

6 MARKING

6.1 Each container shall be legibly and indelibly marked with the following information:

- Name of the product, namely "Packaged Drinking Water";
- Name and address of the manufacturer;
- Batch number;
- Date of packing and "best before" date;

- e) Net volume in ml or litres; and
- f) The words "Synthetic demo label" for this illustrative document.

6.2 Containers shall not be marked with any claim of medicinal or therapeutic property.

7 SAMPLING

7.1 Representative samples of the water shall be drawn as prescribed in DEMO-905.

7.2 The number of containers to be selected from a batch shall be as given in Table 2.

Table 2 Scale of Sampling

Batch Size (Containers)	Number of Containers to be Selected
Up to 5 000	6
5 001 to 25 000	9
25 001 and above	12

7.3 The batch shall be declared conforming if all the test results satisfy the requirements of clause 4.

8 TEST METHODS

8.1 Colour - Colour shall be determined by visual comparison with platinum-cobalt standards.

8.2 Odour - Odour shall be assessed by at least two trained assessors at room temperature.

8.3 Turbidity - Turbidity shall be determined with a nephelometer calibrated with formazin standards.

8.4 Total Dissolved Solids - A filtered sample of 100 ml shall be evaporated at 180 degrees C and the residue weighed.

8.5 pH - pH shall be measured with a pH meter calibrated with two buffer solutions.

8.6 Chemical Parameters - Nitrate, fluoride, chloride and lead should be determined by ion chromatography or atomic absorption spectrometry.

8.7 Microbiological Examination - Coliform bacteria and Escherichia coli shall be tested by the membrane filtration method prescribed in DEMO-905.

ANNEX A

(Clause 4.4.2)

A-1 AUTOMATIC WASHING OF RETURNABLE CONTAINERS

A-1.1 Returnable containers shall be washed with a detergent solution, rinsed with treated water and sanitized before refilling.

A-1.2 Containers showing cracks, scratches or odour shall be rejected.

AMENDMENT NO. 1 JUNE 2026

(Page 2, clause 4.3.3) - Substitute "20 per ml" for "50 per ml". This illustrative amendment demonstrates how BISense records amendments; it does not describe any real revision.